

Clinical Reference Guide: Rectal Cancer

Extracted Focus & Question-Answer Summary from Cancer.gov PDQ

DOCUMENT OVERVIEW

Focus Condition:	Rectal Cancer
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Question 1: What is (are) Rectal Cancer ?

Rectal cancer is a disease in which malignant (cancer) cells form in the tissues of the rectum. The rectum and anal canal make up the last 6 to 8 inches of the large intestine, ending at the anus (the opening to the outside of the body). Along with the esophagus, stomach, small intestine, and colon (the first 5 feet of the large bowel), the rectum is a vital component of the digestive system, which processes nutrients from food and passes waste material out of the body.

Question 2: Who is at risk for Rectal Cancer? ?

An individual's personal and family health history significantly impacts their risk of developing rectal cancer. Older age is a primary risk factor for most cancers, but specific colorectal risk factors include:

- Having a family history of colon or rectal cancer in a first-degree relative (parent, sibling, or child).
- Having a personal history of cancer of the colon, rectum, or ovary.
- Having a personal history of high-risk adenomas (colorectal polyps measuring 1 centimeter or larger, or exhibiting abnormal cellular histology).
- Having inherited genetic modifications that increase the risk of familial adenomatous polyposis (FAP) or Lynch syndrome (hereditary nonpolyposis colorectal cancer).
- Having a personal history of chronic ulcerative colitis or Crohn's disease for 8 years or more.
- Drinking three or more alcoholic beverages per day.
- Smoking cigarettes.
- Being Black or African American.
- Being obese.

Question 3: What are the symptoms of Rectal Cancer ?

Signs and symptoms may be caused by rectal cancer or by benign lower gastrointestinal conditions. Patients should consult a physician if they experience any of the following:

- Blood in the stool, which may appear either bright red or very dark.
- A persistent change in bowel habits, including diarrhea, constipation, or a feeling that the bowel does not empty completely.
- Stools that are narrower or have a different shape than usual.
- General abdominal discomfort, such as frequent gas pains, bloating, fullness, or cramps.
- Change in appetite or unexplained weight loss.
- Feeling very tired or lethargic.

Question 4: How to diagnose Rectal Cancer ?

Specialized examinations of the colon and rectum, along with laboratory tests, are required to diagnose rectal cancer:

- **Physical Exam and Medical History:** Checking general health signs, personal habits, past illnesses, and performing a clinical exam for unusual lumps.
- **Digital Rectal Exam (DRE):** Insertion of a lubricated, gloved finger into the lower rectum (and vagina in women) to feel for lumps or abnormalities.
- **Colonoscopy:** Examination of the entire rectum and colon using a thin, lighted tube with a lens (colonoscope) to identify and remove polyps or tissue samples.
- **Biopsy:** Microscopic examination of extracted cells or tissues by a pathologist to confirm malignancy. Biopsy tissue may also be tested for Lynch syndrome gene mutations via *Reverse transcription-polymerase chain reaction (RT-PCR)* or *Immunohistochemistry* to help guide treatment planning.
- **Carcinoembryonic Antigen (CEA) Assay:** A blood test measuring levels of CEA, a tumor marker that may be elevated in individuals with rectal cancer.

Question 5: What is the outlook for Rectal Cancer ?

The prognosis (probability of recovery) and the selection of appropriate treatment pathways depend on several crucial clinical factors:

- **Disease Stage:** Whether cancer is confined to the inner lining, involves the whole rectal wall, or has spread to lymph nodes, nearby organs, or distant sites.
- **Wall Invasion:** Whether the tumor has spread into or through the bowel wall.
- **Anatomical Location:** Where the tumor is positioned within the rectum.
- **Complications:** Whether the bowel is obstructed or perforated (has a hole).

- **Surgical Resectability:** Whether the tumor can be removed completely by surgery.
- **Patient Profile & History:** The patient's general health and whether the cancer is newly diagnosed or a recurrence.

Question 6: What are the stages of Rectal Cancer ?

Staging uses chest x-rays, CT scans, MRI, PET scans, and *endorectal (transrectal) ultrasound* to find out how far cancer has spread through tissue, the lymphatic system, or the bloodstream. The clinical stages of rectal cancer include:

- **Stage 0 (Carcinoma in Situ):** Abnormal cells are found only in the mucosa (innermost lining) of the rectal wall.
- **Stage I:** Cancer has spread from the mucosa into the submucosa and may have spread into the muscle layer of the rectal wall, but not to lymph nodes.
- **Stage II:** Subdivided into *Stage IIA* (spread through the muscle layer to the outer serosa), *Stage IIB* (spread through the serosa without invading nearby organs), and *Stage IIC* (spread through the serosa into nearby organs). No lymph node involvement.
- **Stage III:** Cancer has involved regional lymph nodes:
 - *Stage IIIA:* Cancer involves the inner lining or muscle layer with 1 to 3 positive lymph nodes (or nearby tissue tumor cells), or involves the submucosa with 4 to 6 positive nodes.
 - *Stage IIIB:* Cancer involves outer layers without organ invasion alongside 1 to 3 positive nodes; or involves inner/muscle layers with 4 to 6 positive nodes; or involves mucosal/muscle layers with 7 or more positive nodes.
 - *Stage IIIC:* Cancer involves outer serosa with 4 to 6 positive nodes; or involves muscle/serosa layers with 7 or more positive nodes; or involves nearby organs with 1 or more positive nodes.
- **Stage IV:** Advanced metastatic disease:
 - *Stage IVA:* Spread to one distant organ not near the rectum (e.g., liver, lung, ovary) or a distant lymph node.
 - *Stage IVB:* Spread to more than one distant organ or into the abdominal wall lining (peritoneum).

Question 7: What are the treatments for Rectal Cancer ?

Management of rectal cancer utilizes five standard modalities, often combined based on stage and anatomical location:

- **Surgery:** The primary treatment for all stages. Techniques include *Polypectomy* or *Local Excision* for early surface lesions; *Resection* of the rectum, adjacent tissues, and lymph nodes for deeper invasion; *Radiofrequency Ablation* or *Cryosurgery* to destroy tissue; and *Pelvic Exenteration* for extensive multi-organ involvement. Reconstructive surgery involves either sewing healthy bowel sections together (anastomosis) or creating a temporary or permanent abdominal opening (colostomy/stoma).

- **Radiation Therapy:** Uses external radiation machines or internal radioactive implants (brachytherapy) to kill cancer cells. Short-course preoperative radiation (fewer, lower doses followed quickly by surgery) is utilized for certain tumor types. Radiation is frequently given before surgery (neoadjuvant) to shrink tumors and improve bowel control, or after surgery (adjuvant) to prevent recurrence.
- **Chemotherapy:** Systemic or regional drugs that halt cancer cell division. For liver metastases, *chemoembolization of the hepatic artery* may be performed to deliver concentrated drugs directly to liver tumors while blocking their blood supply.
- **Active Surveillance:** Close monitoring of patient condition using DRE, MRI, endoscopy, CT scans, and CEA blood tests without immediate treatment, intervening only if test results show cancer progression or recurrence.
- **Targeted Therapy:** Drugs that attack specific cancer cell mechanisms without harming normal cells. Includes monoclonal antibodies (e.g., bevacizumab, ramucirumab, cetuximab, panitumumab) and angiogenesis inhibitors (e.g., ziv-aflibercept, regorafenib) that block tumor blood vessel growth or epidermal growth factor receptors.
- **Clinical Trials & Follow-up:** Ongoing trials test new drug combinations and surgical techniques. Regular medical follow-up, including blood CEA level checks, is mandatory after treatment to monitor recovery and catch recurrences early.

Question 8: What research (or clinical trials) is being done for Rectal Cancer ?

Clinical trials represent a vital part of the cancer research process, designed to determine whether new treatments, medical devices, or surgical procedures are safer and more effective than standard care. Many of today's standard rectal cancer treatments are based on earlier clinical trials.

Patients can enter clinical trials before starting initial treatment, during standard therapy, or after standard regimens have proven ineffective. Participation directly contributes to improving future oncology care and gives patients access to emerging therapeutic modalities.

This document was generated for educational and clinical reference purposes based on official patient treatment guidelines (PDQ) from the National Cancer Institute.